

Ryan Reddy

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Projects

CompanyBrain [🔗](#)

Jul 2026 – Present

Solo AI knowledge-graph platform for internal company documents

- Designed and built, solo, a full-stack platform (Next.js, TypeScript, Prisma, PostgreSQL/pgvector, Vercel) that turns company documents into a source-linked knowledge graph for AI agents that act on company processes.
- Built an LLM extraction pipeline (Claude forced tool use) that pulls entities and relationships against a fixed ontology, tagging each fact with its source document and authority (official policy vs. tribal knowledge).
- Designed two-stage entity resolution (exact match, then Voyage embeddings with pgvector similarity); replaced a guessed 0.85 merge threshold that missed a real duplicate with 0.75, set from measured duplicate vs. non-duplicate scores.
- Added semantic search ranking graph entities by pgvector similarity to a free-text query; verified on live data.
- Capped LLM spend with an in-database ledger and a fail-closed pre-call budget gate.
- Fixed a rate-limit failure found in live testing (batched embedding calls) and three bugs from a code review (27 unit tests); isolated development from production data with a Neon database branch.

Purdue VIP – Low-Power Computer Vision Challenge (LPCVC) Team

Aug 2026 – Present

On-device vision-language model benchmarking and remote-access infrastructure

- Built PhoneBench so teammates can queue benchmark jobs on a shared Snapdragon phone from a website instead of USB/adb: a Kotlin worker app, a FastAPI and Supabase job queue with Google sign-in, and a React site on Vercel. Live since Oct 2026; the first cloud job completed all 100 runs.
- Built a benchmark harness on Qualcomm's GenieX Android sample that runs Qwen3-VL-2B on CPU, GPU, and NPU, recording time-to-first-token, prefill, and decode speed; over 10 runs per configuration, GPU and NPU cut time-to-first-token 2–3× versus CPU (cold start: 30.1 s to 9.6 s).
- Built an accuracy pilot scoring real-vs.-tampered image verdicts against a labeled dataset.

Robot Tour Autonomous Robotics System

2022 – 2024

Lead architect; partner-led code development

- Won 1st place at the Texas State Tournament by architecting a command-based C++/Arduino motion-control stack for a Pololu robot with encoder distance tracking, IMU/gyro yaw estimation, and waypoint/state tracking.
- Designed precision-control logic (PID-style correction, S-curve acceleration, drift correction, turn calibration, motor balancing) and a custom PCB integrating gyroscope and Hall effect encoder feedback for repeatable autonomous runs.

Education

Purdue University

Expected May 2029

B.S. in Computer Science

- GPA: 3.44/4.0

Awards and Honors

- Top 30 USA PicoCTF 2024; AIME Qualifier 2024; CyberPatriot Nationals Semifinalist 2024
- 1st Texas State Tournament Science Olympiad Robot Tour 2024; 2nd Texas State Tournament Science Olympiad Solar Power 2024; FBLA Nationals Qualifier 2024

Technologies

Languages: Python, TypeScript, JavaScript, Kotlin, C++, C#, Java, SQL, HTML, CSS, Bash

Web/Backend: React, Next.js, Node.js, FastAPI, Prisma, PostgreSQL/pgvector, Supabase, Neon, Vercel, Tailwind CSS, REST APIs, pytest, Git

AI/Systems: Anthropic API (tool use), Voyage AI, embeddings and vector search, knowledge graphs, Claude Code, OpenAI Codex, Android/adb, matplotlib, Arduino, Unity, Logisim, NgSpice, TensorFlow